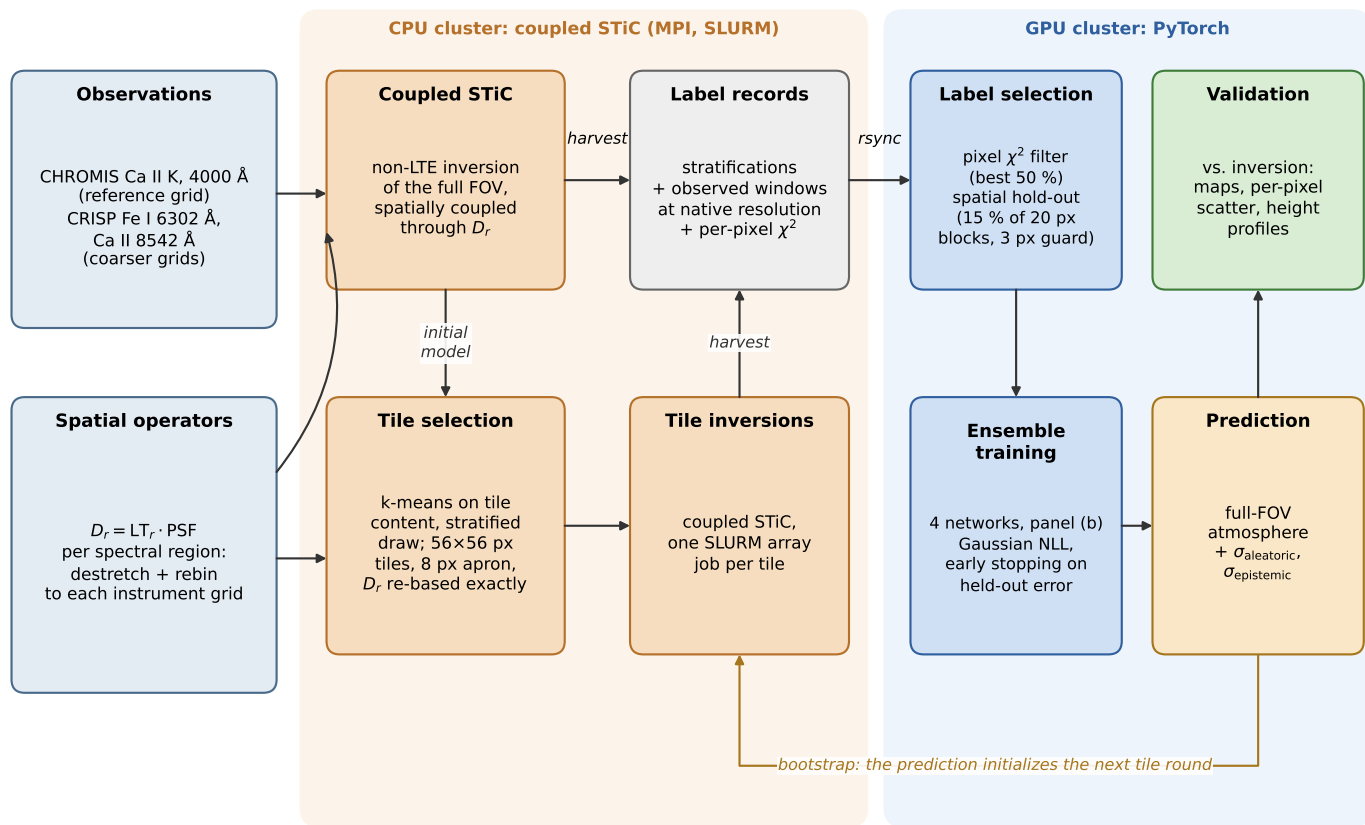
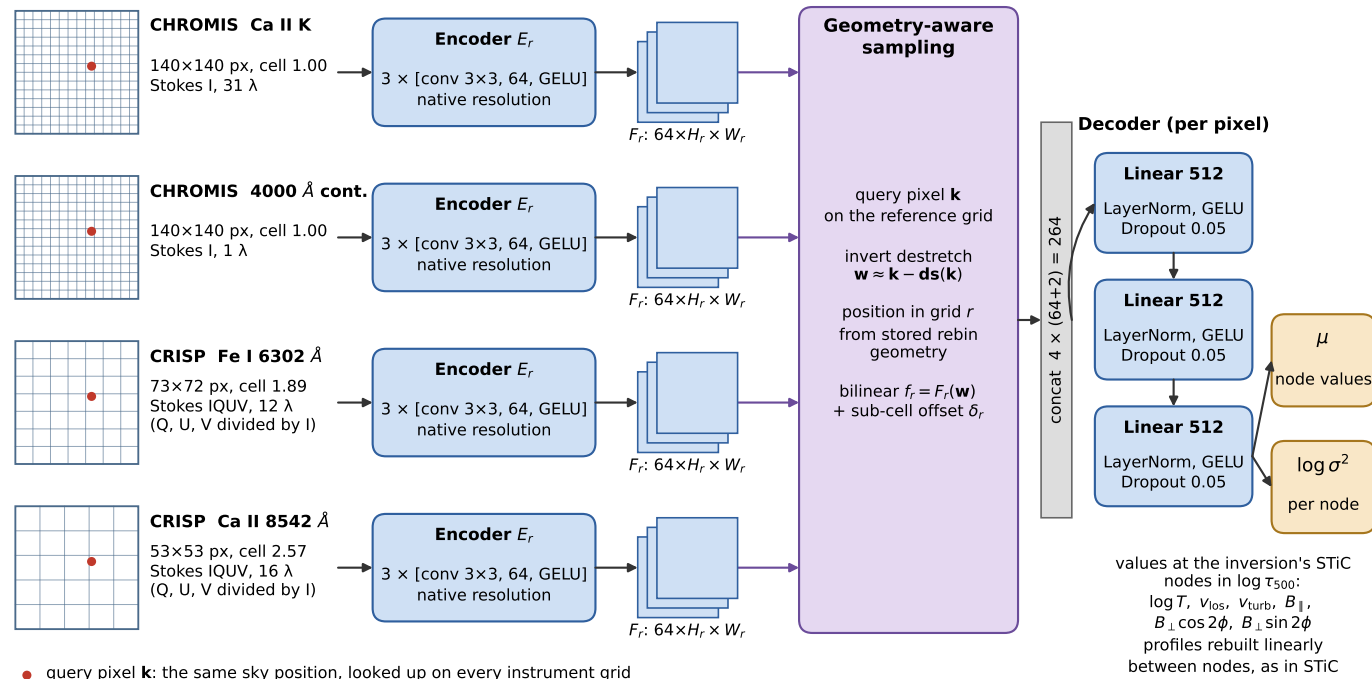


(a) Label generation, training and the bootstrap loop



(b) Multi-resolution emulator network (one ensemble member)



• query pixel $\mathbf{k}$: the same sky position, looked up on every instrument grid

Training and uncertainty

Loss: Gaussian negative log-likelihood $\frac{1}{2}[\log \sigma^2 + (y - \mu)^2/\sigma^2]$ on standardized targets, plus $\lambda \sum (\partial^2 \mu / \partial \log \tau^2)^2$ with $\lambda = 10^{-3}$
AdamW (lr 3×10^{-4} , cosine schedule), $\leq 20\,000$ steps of 4 records $\times \leq 2048$ pixels, early stopping on the held-out prediction error
Ensemble of 4 members (different initializations): prediction = ensemble mean, $\sigma_{\text{aleatoric}} = \{\sigma^2\}^{1/2}$, $\sigma_{\text{epistemic}}$ = spread of the member means